

AL-2002 - Artificial Intelligence Lab

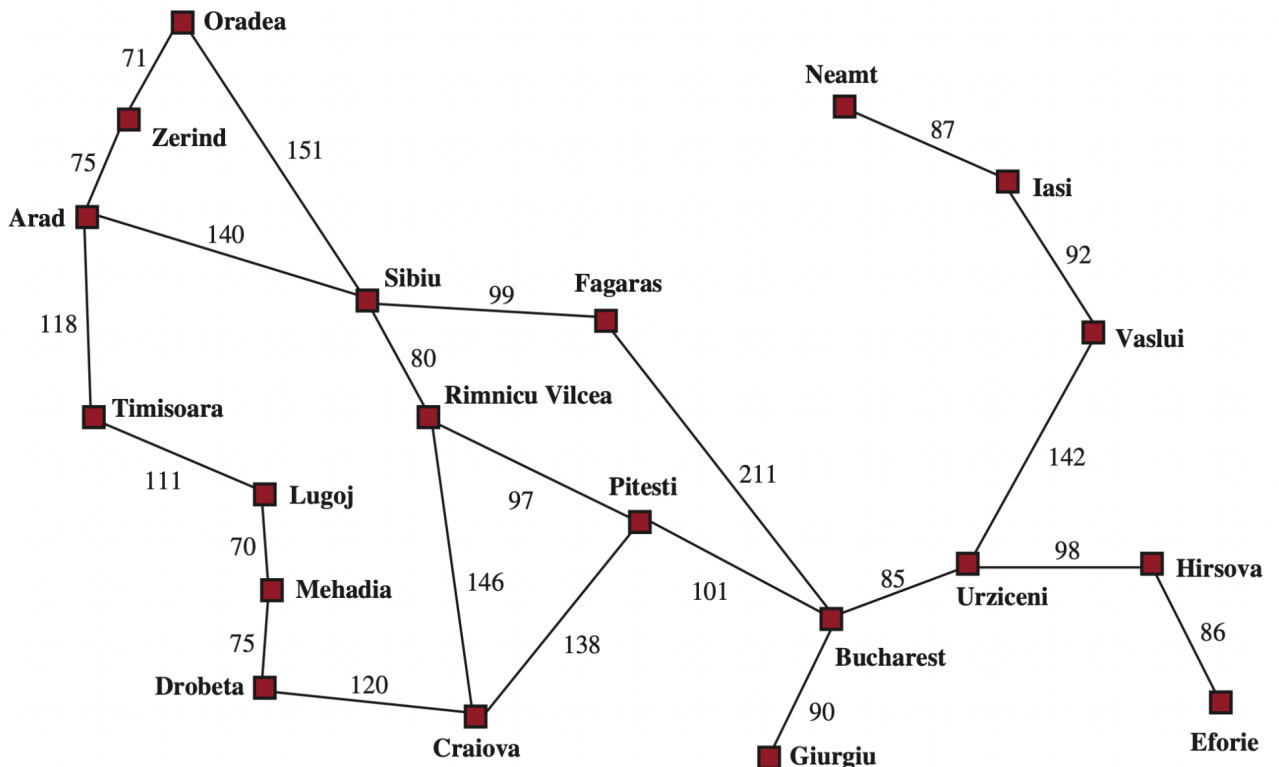
Lab # 3

Objectives:

- Uninformed Search (Bidirectional Search + UCS) + Informed Search (Best First Search)

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook.
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. **Do not copy from any source otherwise you will be penalized with negative marks.**
5. Add your source code in word document and attach output screenshots.
6. Upload word file and ipynb file.
7. Please submit your **Both files** with this naming convention **ROLLNO_9A_LABNO. (Do not upload zip file).**
8. Submit your lab on Google Classroom.



Question 1:

A road rescue coordination system needs to locate a stranded traveler somewhere between Arad and Bucharest. To save time, two rescue teams are deployed simultaneously one starting from Arad and one starting from Bucharest each exploring roads outward until they meet somewhere in the middle. Using the Romania road map, implement **Bidirectional Search** where both searches expand level by level (BFS from both ends). The program should display which cities each team explores at each step, identify the meeting point, and reconstruct the full path from Arad to Bucharest through that meeting point. Also display the total number of cities explored compared to a standard BFS to show the efficiency gain.

Question 2:

A logistics company is developing an intelligent route planning system to transport goods between major cities at the lowest possible cost. Each city is represented as a node in a graph, and each road connecting two cities is represented as an edge with an associated travel cost (such as fuel cost or distance). Since different routes have different costs, the system must determine the least-cost path rather than the path with the fewest stops. To achieve this, the system applies Uniform Cost Search (UCS), which always expands the node with the lowest cumulative path cost. Write a Python program to implement Uniform Cost Search (UCS) to find the minimum-cost path between a given source city and a destination city. Your program should display the path taken and the total travel cost required to reach the destination.

Question 3:

A GPS navigation system needs to find a route from any starting point to Bucharest as efficiently as possible. Unlike BFS and DFS which explore blindly, this system uses an **estimated distance to the destination** to decide which city to explore next, always prioritizing the city that appears closest to Bucharest.

Using the Romania road map, implement **Best First Search (Greedy)** with the following steps:

Step 1: Calculate Heuristic Table:

Before writing any code, manually trace the most direct route from every city to Bucharest using the road distances on the given map. Use that total distance as your heuristic estimate $h(n)$ for that city. Store all values in a dictionary where the key is the city name and the value is your estimated distance to Bucharest. Note that $h(\text{Bucharest})$ must always be 0.

Step 2: Implement Best First Search:

Using your heuristic dictionary, implement Best First Search. The algorithm must always explore the unvisited city with the **lowest $h(n)$ value** next. Display which city is being explored at each step along with its heuristic value.